PraK Tool V3: Enhancing Video Item Search Using Localized Text and Texture Queries

Michael Stroh[1(✉)], Vojtěch Kloda[2], Benjamin Verner[2], Zuzana Vopálková[2], Raphael Buchmüller[3], Bastian Jäckl[3], Jakub Hajko[2], and Jakub Lokoč[2]

[1] Visual Computing, Department of Computer Science Faculty of Natural Sciences, Konstanz University, Konstanz, Germany
michael.stroh@uni-konstanz.de

[2] SIRET Research Group, Department of Software Engineering Faculty of Mathematics and Physics, Charles University, Prague, Czech Republic
jakub.lokoc@matfyz.cuni.cz

[3] Data Analysis and Visualization, Department of Computer Science Faculty of Natural Sciences, Konstanz University, Konstanz, Germany
{raphael.buchmueller,bastian.jaeckl}@uni-konstanz.de

Abstract. We present a third version of the PraK system designed around an effective text-image and image-image search model. The system integrates sub-image search options for localized context search for CLIP and image color/texture and interfaces tailored to specific task types. A user study shows the benefits of sub-image search in homogeneous video collections. We further enhance the user experience of the PraK framework by re-designing the frontend and introducing task-specific views.

Keywords: interactive video retrieval · deep features · text-image retrieval

1 Introduction

The Video Browser Showdown (VBS) competition [2,4,12] is an annual event that defines challenging video search problems. While the first iterations of VBS focused on effective browsing in a single video file, later iterations allowed arbitrary retrieval mechanisms in extensive video archives (except using cameras). With the advancements of deep neural networks allowing effective text-image search [3], the VBS task pool started to include also highly challenging homogeneous collections such as Marine Video Kit [11] and Medical videos. Unlike heterogeneous datasets such as V3C [8], the new homogeneous datasets represent a tough challenge even for smaller collections. The reason is that text search often does not help due to high visual and concept redundancy. Yet, every scene in the homogeneous datasets is unique, and thus, novel retrieval mechanisms are necessary to aid users with searching.

I. Ide et al. (Eds.): MMM 2025, LNCS 15524, pp. 326–333, 2025.
https://doi.org/10.1007/978-981-96-2074-6_39

Based on the results of VBS 2024, the two most effective systems (Visione [1] and Vibro [10]) were successful by making use of state-of-the-art deep neural networks (based on the CLIP architecture [7]) combined with intelligent user interfaces for browsing and query refinement. Specifically, Visione often relied on late fusion of multiple cross-modal models and temporal queries. The Vibro system relied on text-based CLIP models and an effective image-to-image similarity model. Hence, the third version of our system is designed (again) around an effective CLIP version and interactive search approaches. Specifically, the third version of the PraK system [6] is enhanced in several ways. First, the system supports the highly effective deep neural network model [9] provided by the Vibro team. According to our experiments, the new embeddings alone substantially improve search effectiveness. Second, for inhomogeneous collections, we re-introduce free-form text searching in selected image sub-regions [5]. Although this technique increases demands on storage resources and requires more time to pre-process (more features need to be extracted and stored), it provides a more effective ranking of selected keyframes in homogeneous data. Furthermore, this method does not increase the computational complexity during query time, thus allowing for more specialized queries and system interactions. Third, the new version also allows query refinement by image sub-regions and their color-texture features.

2 PraK System Architecture and Description

The PraK system [6,13] is a framework with three key components working together. First, the Video Data Service acts as a stateless in-memory database, providing quick access to data and embeddings from the corresponding model upon query. It integrates with a stateful Application Service Backend, which connects the Frontend to the Video Data Service. It performs essential data transformations, such as post-processing or user-specified adjustments like Bayesian Relevance Feedback for ordering video items. Finally, the Web Frontend offers an accessible interface, allowing users to perform retrieval tasks independently. Together, these components form a robust system for efficiently managing and displaying video or image data.

This architecture of separated responsibility allowed the system to use powerful search methodologies and query tools while scaling to multiple concurrent users. PraK uses text-to-image similarity queries and image-to-image similarity search based on the joint embedding space of CLIP [3] models, Bayesian relevance feedback for item re-ranking, temporal queries that enable users to search for different scenes appearing sequentially within the same video, and a user-specific history cache that allows user to undo unsuccessful operations quickly. On overview of PraK's architecture and used technologies is illustrated in Fig. 1.

In this work, we further leverage the available foundation of the PraK framework [6] by primarily introducing new grid-based query capabilities to enable users to perform more localized text and image similarity searches. The effectiveness, especially on homogenous image datasets, is described in Sect. 4.

By reworking the Frontend, we further emphasized usage efficiency while maintaining an easily understandable interface. We used tooltips, different query-element layouts, and different data viewing modes based on various tasks like KIS and AVS to focus on the more effective search mechanisms based on PraK's experiences from previous iterations of the competition, which are described further in Sect. 3.

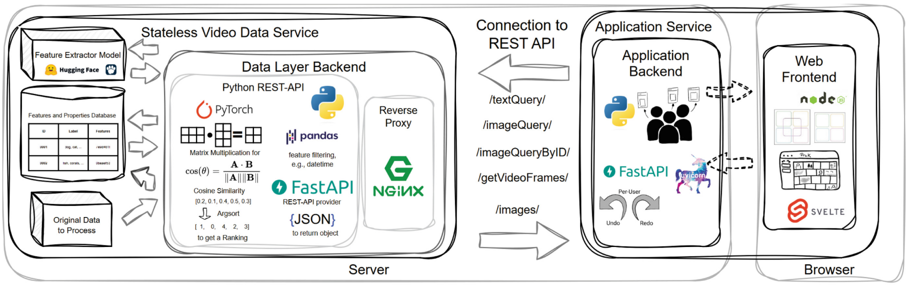

Fig. 1. Architecture of the presented video search system based on stateless Video Data Services [13]. The Application Backend handles Bayesian re-ranking and state for various users, while the Web Frontend allows interaction with the system.

3 Added Features for VBS 2025

3.1 Reworked User Interface for Different Tasks

The new layout of different frontend elements allows a more intuitive view of the available query methods. It leaves the maximum space for displaying the ranked query images as large as possible. We introduce views for each of the three tasks Q&A, AVS and KIS that move different layout elements for a more effective search experience depending on the dataset and user task as displayed in Fig. 1. For example, for the V3C dataset, we would favor the CLIP-driven text queries while enabling the introduced color descriptor image-patch query for the homogeneous marine and medical datasets. These features are further described in Sect. 3.

We further introduce a new video exploration view that will display one selected video with the different chosen keyframes from the video to explore temporal video context effectively and quickly, as well as select time ranges or individual frames to submit as solutions to a task.

Wherever possible, we include visual indicators that show the results of submitted queries based on feedback from the submission evaluation server used in the completion to indicate successful or unsuccessful submissions. Figure 2 shows the modified menus for the different tasks and an illustrative depiction of an image patch selection.

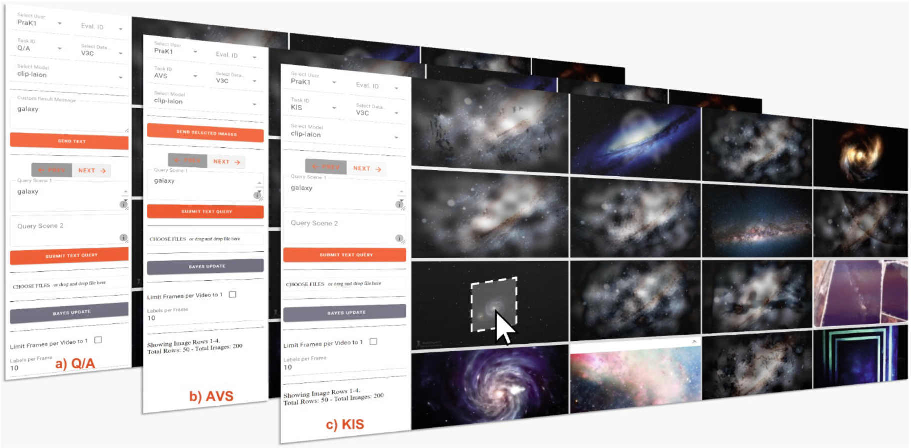

Fig. 2. Prototype of the new menu layouts for the PraK frontend with task-dependent interface variations for a) Q&A, b) AVS and c) KIS.

3.2 Localized Clip Feature Queries

While text-to-image-based similarity has become one of the most effective methods for finding specific images, there is no practical way to localize different semantic objects within an image using a fixed feature vector for an image if initial search queries containing a localization term in text-form fail to produce the correct image rankings. We introduce text-to-image queries within a selected image region to mitigate this limitation of the current embedding models. Performing the text queries not on the whole image but on a sub-region enables users to limit the search context and perform more specific queries, as they potentially avoid describing false positives in different regions of the whole image.

The image partitioning we use in this work is based on a fixed grid with overlapping grid cells. We reasoned our choice of grid with a study we conducted on the VBS-related datasets in Sect. 4. We pre-compute all CLIP feature vectors for all image partitions on the backend server, stored such that for a localized query, the system is only required to compute the feature similarity of the query and the feature vectors corresponding to image patches of the given location. This structure maintains fast query times in favor of smaller storage requirements.

3.3 Patch-Based Image Texture Queries

Similar to the CLIP text-based search, we additionally introduce an image-patch query method that allows users to perform an image-to-image similarity based on small but distinctive regions to find their target image.

Using color and image texture descriptors as features for these image patches, we enable users to search based on purely visual and highly localized aspects of specific images in their current display. For our application, texture refers to the visual patterns or spatial arrangement of pixel intensities in an image that can represent specific image objects or scene structures. These texture descriptors can be based on patch statistics, like intensity histograms, frequency-based, like those derived from wavelet- or Fourier transform, or derived from Gabor filters. These descriptors can then be encoded and the similarities of these patch texture descriptors can be used to query for visual image sub-structures. This can be very useful for homogenous datasets, where semantic objects are sparse, generic, or repeated in minor variations throughout the dataset. In these cases, the classic text-to-image querying strategy often fails, as minute color details are not captured well within the CLIP latent space, and sceneries are challenging to describe distinctively using a textual description.

Instead of a large partition with few useful localizations, we opted for a fine grid-based partition of the image so users can select the specific colors and texture that potentially lead to a more promising image re-ranking.

4 User Study for Grid-Based Image-Subregion Search

A new feature of the PraK system is querying in image subregions. When querying the system, the user can select one of several pre-determined subregions where the query should be executed. Doing so will enact the query in the matching subregions in the image database, limiting the context of the original image.

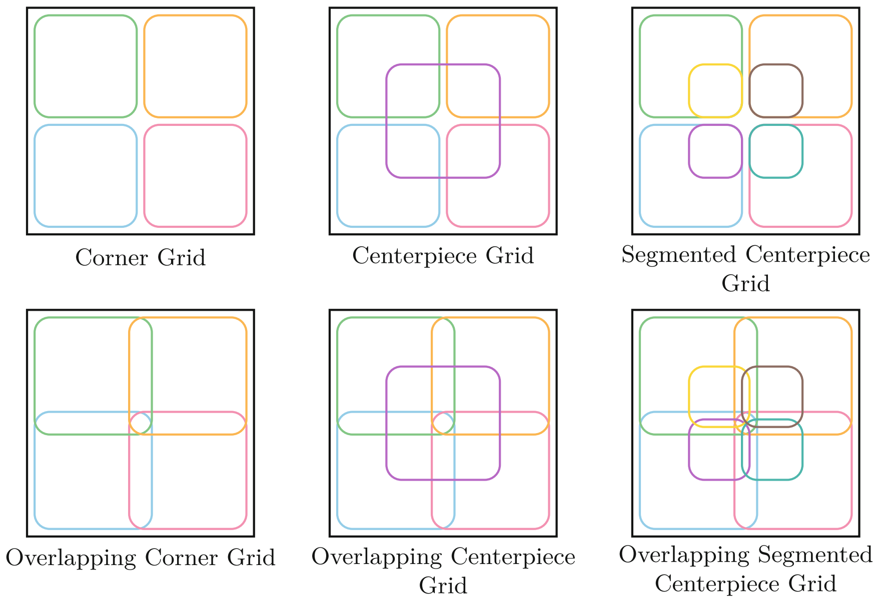

Fig. 3. Schemas of the different grid layouts.

Six distinct grid designs were proposed and evaluated. The corner grid is made by splitting the original image in half in both dimensions. The centerpiece grid is a corner grid with an additional fifth rectangle placed in the middle. The centerpiece rectangle has equal dimensions to the corner rectangles. The segmented centerpiece grid also splits the centerpiece grid's centerpiece rectangle into four equal rectangles, similar to the corner grid.

The last three grids were made by elongating the sides of the rectangles of the first three grids by a constant factor. This causes a significant overlap between the rectangles. The different grid layouts are illustrated in Fig. 3.

A user study was conducted to test the effectiveness of the different grids. Users were given a series of random images taken from the image database and tasked with finding and annotating interesting objects. To do so, the user had to draw a rectangle around the object of interest and write a textual description. The final set of annotations was then used to test the grids' effectiveness automatically.

The evaluation aims to simulate a real user using the system. For a given annotated object, the evaluation suite selects the image subregion with the most prominent intersection with the object. It uses the textual description as a query, yielding a series of all database images ranked by relevance. The position of the image containing the object in this series is used as a performance indicator. The suite's final result is the average position of all annotated objects and the count of objects that ranked in the top k positions for a given k.

The table shown in Fig. 4 below highlights the results of one model with 101 user annotations.

Average Positions							
	no grid	corner	corner overlap	centerpiece	centerpiece overlap	segmented centerpiece	segmented centerpiece overlap
MCIP-ViT-SO400M-14-SigLIP-384	1520	902	772	691	682	772	1090

Top-100 Positions of 101 Annotations							
	no grid	corner	corner overlap	centerpiece	centerpiece overlap	segmented centerpiece	segmented centerpiece overlap
MCIP-ViT-SO400M-14-SigLIP-384	46	53	56	59	57	56	51

Fig. 4. User study results for various grid settings.

In all cases, the overlapping centerpiece grid outperformed all the others, including the case where no grid system was used. The segmented centerpiece grid performed identically to the corner overlap grid due to the nature of the image subregion selection algorithm; the centerpiece subregions were subsets of the corner subregions, and because the corner ones had precedence, the centerpiece subregions were never selected. An alternative algorithm preferring smaller image subregions by considering the relative area overlap instead of the absolute one was proposed and tested; the results were considerably worse in most cases.

5 Conclusions

The third version of the PraK system offers significant improvements text-image and image-image search modalities. By integrating sub-image search options for localized context search within CLIP, as well as image color and texture query capabilities, the system demonstrates enhanced utility across various tasks. A user study confirms the value of these advancements, particularly in supporting sub-image searches within homogeneous video datasets.

References

1. Amato, G., et al.: VISIONE 5.0: Enhanced user interface and AI models for VBS2024. In: Rudinac, S., Hanjalic, A., Liem, C.C.S., Worring, M., Jónsson, B., Liu, B., Yamakata, Y. (eds.) MultiMedia Modeling - 30th International Conference, MMM 2024, Amsterdam, The Netherlands, January 29 - February 2, 2024, Proceedings, Part IV. Lecture Notes in Computer Science, vol. 14557, pp. 332–339. Springer (2024). https://doi.org/10.1007/978-3-031-53302-0_29
2. Heller, S., et al.: Interactive video retrieval evaluation at a distance: comparing sixteen interactive video search systems in a remote setting at the 10th video browser showdown. Int. J. Multim. Inf. Retr. **11**(1), 1–18 (2022). https://doi.org/10.1007/S13735-021-00225-2
3. Ilharco, G., et al.: Openclip (2021)
4. Lokoc, J., et al.: Interactive video retrieval in the age of effective joint embedding deep models: lessons from the 11th VBS. Multim. Syst. **29**(6), 3481–3504 (2023). https://doi.org/10.1007/S00530-023-01143-5
5. Lokoc, J., Mejzlík, F., Veselý, P., Soucek, T.: Enhanced somhunter for known-item search in lifelog data. In: Gurrin, C., Schoeffmann, K., Jónsson, B., Dang-Nguyen, D., Lokoc, J., Tran, M., Hürst, W., Rossetto, L., Healy, G. (eds.) Proceedings of the 4th Annual on Lifelog Search Challenge, LSC@ICMR 2021, Taipei, Taiwan, 21 August 2021, pp. 71–73. ACM (2021). https://doi.org/10.1145/3463948.3469074
6. Lokoc, J., Vopálková, Z., Stroh, M., Buchmueller, R., Schlegel, U.: Prak tool: An interactive search tool based on video data services. In: Rudinac, S., Hanjalic, A., Liem, C.C.S., Worring, M., Jónsson, B., Liu, B., Yamakata, Y. (eds.) MultiMedia Modeling - 30th International Conference, MMM 2024, Amsterdam, The Netherlands, January 29 - February 2, 2024, Proceedings, Part IV. Lecture Notes in Computer Science, vol. 14557, pp. 340–346. Springer (2024). https://doi.org/10.1007/978-3-031-53302-0_30
7. Radford, A., et al.: Learning transferable visual models from natural language supervision. In: Meila, M., Zhang, T. (eds.) Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event. Proceedings of Machine Learning Research, vol. 139, pp. 8748–8763. PMLR (2021). http://proceedings.mlr.press/v139/radford21a.html
8. Rossetto, L., Schuldt, H., Awad, G., Butt, A.A.: V3C - A research video collection. In: International Conference on Multimedia Modeling, pp. 349–360. Springer (2019https://doi.org/10.1007/978-3-030-05710-7_29
9. Schall, K., Barthel, K.U., Hezel, N., Jung, K.: Optimizing CLIP models for image retrieval with maintained joint-embedding alignment. CoRR **abs/2409.01936** (2024). https://doi.org/10.48550/ARXIV.2409.01936, https://doi.org/10.48550/arXiv.2409.01936

10. Schall, K., Hezel, N., Barthel, K.U., Jung, K.: Optimizing the interactive video retrieval tool vibro for the video browser showdown 2024. In: Rudinac, S., Hanjalic, A., Liem, C.C.S., Worring, M., Jónsson, B., Liu, B., Yamakata, Y. (eds.) MultiMedia Modeling - 30th International Conference, MMM 2024, Amsterdam, The Netherlands, January 29 - February 2, 2024, Proceedings, Part IV. Lecture Notes in Computer Science, vol. 14557, pp. 364–371. Springer (2024https://doi.org/10.1007/978-3-031-53302-0_33
11. Truong, Q.T., Vu, T.A., Ha, T.S., Lokoč, J., Tim, Y.H.W., Joneja, A., Yeung, S.K.: Marine video kit: A new marine video dataset for content-based analysis and retrieval. In: MultiMedia Modeling - 29th International Conference, MMM 2023, Bergen, Norway, January 9-12, 2023. Springer (2023). https://doi.org/10.1007/978-3-031-27077-2_42
12. Vadicamo, L., et al.: Evaluating performance and trends in interactive video retrieval: Insights from the 12th VBS competition. IEEE Access **12**, 79342–79366 (2024). https://doi.org/10.1109/ACCESS.2024.3405638
13. Vopálková, Z., Yaghob, J., Stroh, M., Schlegel, U., Lokoc, J.: Searching temporally distant activities in lifelog data with prak tool V2. In: Proceedings of the 7th Annual ACM Workshop on the Lifelog Search Challenge, LSC 2024, Phuket, Thailand, 10 June 2024, pp. 111–116. ACM (2024). https://doi.org/10.1145/3643489.3661131